Weighted single-level versus multilevel estimators under unequal probabilities of selection: a replication of Koziol (2015)

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Abstract

We partly replicated the headline result of Koziol (2015). She reported, in that dissertation (Koziol, 2015), that weighted single-level estimators tended to outperform weighted multilevel estimators, and that multilevel weight approximations did not perform well when selection was informative. On the normal random-intercept cell with an informative design, intraclass correlation .05, and average cluster size 5, she reported intercept biases of 0.000, 0.151, and 0.308 for weighted single-level estimation with sandwich standard errors, weighted multilevel pseudo-maximum likelihood with the true weights, and the same multilevel estimator with approximate cluster weights. Our means, from 998 of 1,000 replications, are 0.001, 0.153, and 0.309. The signed gap, multilevel bias minus single-level bias, is 0.152 against the reported 0.151, in the same direction, and coverage is 0.945 against 0.946 and 0.000 against 0.001. The source software is Mplus 7; ours is R. The tolerance on a difference of two three-decimal biases is 0.001, and a positive difference means our Monte Carlo mean is larger than the printed value. Descriptive point estimates; not equivalence. The Bernoulli outcome and the random-coefficient model were not rerun, so those arms are not established. No sign reversed on the normal random-intercept grid. The largest coverage gap on that grid is the weighted multilevel estimator at intraclass correlation .05 and average cluster size 20: she reported 0.282 and we obtained 0.248.

Estimand	Published	Ours	Effect magnitude	Direction	Verdict
Multilevel minus single-level intercept bias, informative, $\rho = .05$, $\bar{n}_j = 5$ (outcome units; tolerance 0.001)	0.151	0.152	larger	same direction	close
Approximate-weight minus true-weight multilevel intercept bias, same cell (outcome units; tolerance 0.001)	0.157	0.156	same	same direction	match
Single-level minus multilevel intercept coverage, same cell (proportion; tolerance 0.001)	0.945	0.945	same	same direction	match
Bernoulli informative intercept bias, multilevel minus single-level, $\rho = .05$, $\bar{n}_j = 5$	−0.170	not established	not established	not established	not established

Descriptive point estimates; not equivalence. Positive bias contrasts mean the first estimator has the larger positive intercept bias. The Bernoulli contrast is in probit units. Published values are the Mplus 7 Monte Carlo means in Tables C.2 and E.2. Ours are R means. A positive ours-minus-printed gap means our mean is larger.

1 Introduction

We partly replicated the headline result of Koziol (2015): under informative two-stage sampling and a normal random-intercept outcome, weighted single-level estimation with sandwich standard errors recovered the intercept, weighted multilevel pseudo-maximum likelihood did not, and approximate cluster weights were worse than the true weights. She reported intercept biases of 0.000, 0.151, and 0.308 at intraclass correlation .05 and average cluster size 5; our means are 0.001, 0.153, and 0.309. The Bernoulli outcome and the random-coefficient model were not rerun. No sign reversed on the normal random-intercept grid. The tolerance is 0.001 on a difference of two biases printed to three decimals, and the comparison direction on this page is ours minus the printed Mplus value.

Koziol's dissertation, chaired by James A. Bovaird at the University of Nebraska–Lincoln in May 2015, asks three questions (Koziol, 2015, pp. 58–59). How do unweighted single-level and multilevel estimators compare? How do the weighted versions compare? How do multilevel estimators that replace the cluster weight by the population-to-sample cluster ratio compare with estimators that use the true weights? The factors are the population model, informativeness, the outcome distribution, the intraclass correlation, average cluster size, and the estimator (Table 1.1 and Table 3.1). The full cross is 208 settings, each with 1,000 replications. This note reruns the eight normal random-intercept settings, which are the complete factorial of design, intraclass correlation, and cluster size for that model, and leaves the other 200 settings unrun.

2 What the normal random-intercept grid shows

The question is whether weighting repairs the intercept, and whether multilevel pseudo-maximum likelihood repairs it as well as a weighted single-level fit. On every informative normal cell the answer matches the dissertation: the unweighted intercept is badly biased, the weighted single-level intercept is nearly unbiased with coverage near .95, the weighted multilevel intercept is less biased than the unweighted estimator but still biased when clusters are small, and the approximate-weight multilevel intercept is worse than the true-weight multilevel intercept.

Table 1 gives the intercept bias. At the primary cell the unweighted bias is 0.682 against the reported 0.681, the weighted single-level bias is 0.001 against 0.000, the true-weight multilevel bias is 0.153 against 0.151, and the approximate-weight bias is 0.309 against 0.308. Root mean square error of the fixed predictor, her equation (40), is 0.046 against 0.047, 0.156 against 0.155, and 0.310 against 0.309 for those three weighted estimators (Table F.1). Mean squared error of the intercept is 0.0009, 0.0240, and 0.0958 against 0.0009, 0.0235, and 0.0954 (Table D.2). Coverage of the nominal 95 percent interval is 0.945 against 0.946 for the weighted single-level estimator and 0.000 against 0.001 for the true-weight multilevel estimator (Table E.2).

The same order holds at the other three informative cells. At intraclass correlation .25 and average cluster size 5 the biases are 0.919, 0.003, 0.163, and 0.556 against 0.920, 0.005, 0.164, and 0.558. At average cluster size 20 and intraclass correlation .05 they are 0.683, 0.001, 0.045, and 0.201 against 0.681, 0.000, 0.043, and 0.200. At average cluster size 20 and intraclass correlation .25 they are 0.920, 0.002, 0.047, and 0.440 against 0.919, 0.001, 0.046, and 0.440. Larger clusters shrink the true-weight multilevel bias, as she reported, and they do not rescue the approximate weights. Under non-informative sampling every intercept bias rounds to 0.000 or to one thousandth, which is the pattern in Table C.1.

The paired contrast of multilevel bias minus single-level bias at the primary cell has Monte Carlo mean 0.152031 and standard error 0.000416. The printed contrast is 0.151. The gap of about one thousandth is just outside a two-standard-error interval that treats 0.151 as exact, and it sits inside that interval if the printed number is allowed the half-thousandth from which a three-decimal print is taken. The

approximate-weight contrast, 0.156 against 0.157, is inside both the rounding tolerance and two standard errors (standard error 0.000515).

Table 1: Intercept bias for the normal random-intercept model. Published values are from Tables C.1 and C.2. Our values are Monte Carlo means, rounded half up to three decimals. ρ is the intraclass correlation and $\bar{n}_j$ is the target average cluster size.

Design	Cell	us.ics	um.ccs	ws.ics	wtm.ccs	wam.ccs
Informative	$\rho = .05$, $\bar{n}_j = 5$, published	0.681	0.681	0.000	0.151	0.308
Informative	$\rho = .05$, $\bar{n}_j = 5$, ours	0.682	0.682	0.001	0.153	0.309
Informative	$\rho = .25$, $\bar{n}_j = 5$, published	0.920	0.920	0.005	0.164	0.558
Informative	$\rho = .25$, $\bar{n}_j = 5$, ours	0.919	0.919	0.003	0.163	0.556
Informative	$\rho = .05$, $\bar{n}_j = 20$, published	0.681	0.681	0.000	0.043	0.200
Informative	$\rho = .05$, $\bar{n}_j = 20$, ours	0.683	0.683	0.001	0.045	0.201
Informative	$\rho = .25$, $\bar{n}_j = 20$, published	0.919	0.919	0.001	0.046	0.440
Informative	$\rho = .25$, $\bar{n}_j = 20$, ours	0.920	0.920	0.002	0.047	0.440
Non-informative	four cells, published	each printed bias is between -0.002 and 0.001				
Non-informative	four cells, ours	each mean is between -0.002 and 0.002				

Table 2 is the informative coverage comparison. The qualitative conclusion is the same in every cell: weighted single-level coverage stays near .95, approximate-weight coverage of the intercept is 0, and true-weight multilevel coverage is near 0 when clusters are small and rises when clusters are larger. The one numerical mismatch at the third decimal that changes the reading of a cell is true-weight multilevel coverage at $\rho = .05$ and $\bar{n}_j = 20$: 0.248 against 0.282. Both numbers are far below .95, so the conclusion that the interval undercovers is unchanged. A five percent increase in our standard errors would move that coverage to 0.288. Our standard errors match WeMix cluster-robust standard errors on the checked replicate, so the open candidate is a difference between that CR-0 sandwich and the Mplus MLR scaling, together with ordinary Monte Carlo error. Her original Monte Carlo standard error for a coverage near .28 with 1,000 draws is about 0.014; ours is about 0.014. The gap of 0.034 is inside two standard errors of a difference that uses both of those uncertainties, and just outside two standard errors that use only ours.

Table 2: Coverage of the intercept interval under informative sampling, normal random-intercept model. Published values are from Table E.2. Gray in her table marks coverage outside (.932, .968).

Cell	Source	ws.ics	wtm.ccs	wam.ccs	us.ics
$\rho = .05$, $\bar{n}_j = 5$	published	0.946	0.001	0.000	0.000
$\rho = .05$, $\bar{n}_j = 5$	ours	0.945	0.000	0.000	0.000
$\rho = .25$, $\bar{n}_j = 5$	published	0.938	0.016	0.000	0.000
$\rho = .25$, $\bar{n}_j = 5$	ours	0.936	0.019	0.000	0.000
$\rho = .05$, $\bar{n}_j = 20$	published	0.929	0.282	0.000	0.000
$\rho = .05$, $\bar{n}_j = 20$	ours	0.933	0.248	0.000	0.000
$\rho = .25$, $\bar{n}_j = 20$	published	0.939	0.658	0.000	0.000
$\rho = .25$, $\bar{n}_j = 20$	ours	0.947	0.646	0.000	0.000

Slope and level-2 coefficients are a different story, and it matches hers. Under the informative design the level-1 slope bias is a few thousandths for every weighted estimator. Relative bias of γ_{10} at the primary cell is 0.001 for the weighted single-level estimator. Weighting is doing its work on the intercept, which

is the parameter tied to the random intercept that enters the selection probability.

3 What we did ourselves, and what we took from others

Koziol specified the population, the Poisson inclusion probabilities, the eight estimators, and the bias, relative bias, mean squared error, coverage, and root mean square error formulas (Koziol, 2015, ch. 3 and apps. A and C–F). The printed tables are hers. We rewrote the generator from Appendix A, fit the normal random-intercept models, and compared the Monte Carlo means with those tables.

The generator is `simulation/generate.R:7`. An independent Python transcription, `simulation/generate_py.py:1` was given the same random numbers on a reduced population of 400 clusters and returned the same sample size, the same cluster count, and the same mean outcome to machine precision. The production draws use her constants: 48,450 clusters, stage-1 intercept 4.12, stage-2 intercept 1.53, and Method 2 scaling of the conditional weights (Koziol, 2015, eq. 15). On the primary cell the mean number of sampled clusters was 968 and the mean sample size was 4,884, against her Table 4.1 averages of 964 and 4,867. Mean cluster weight was 49.5 with mean standard deviation 38.8, against her 50 and 39.

Estimation followed the analysis section on pages 76–79. Single-level fits are Gaussian survey regression with cluster-robust standard errors, the R analogue of `TYPE=COMPLEX` and `ESTIMATOR=MLR` (Lumley, 2004; Binder, 1983). Unweighted multilevel fits are maximum likelihood linear mixed models (Bates et al., 2015). Weighted multilevel fits are the Gaussian pseudo-likelihood in which the conditional normal density is raised to the Method 2 weight and the cluster contribution is raised to the unscaled level-2 weight (Pfeffermann et al., 1998; Asparouhov, 2006; Rabe-Hesketh and Skrondal, 2006). Koziol names this estimator as implemented in Mplus 7 (Muthén and Muthén, 2012). The approximate cluster weight is the population number of clusters divided by the sample number of clusters (Kovačević and Rai, 2003). The implementation is `simulation/estimators.R:9`. On replication 1 of the primary cell, that fit and WeMix 4.0.3 agreed to about 10^{-6} on the three fixed coefficients (Bailey et al., 2023). The replicate is `simulation/results/wemix_crosscheck_cell1_rep1.csv`. On one further informative draw, `geeglm` with independence working correlation and the unconditional weight matched the survey coefficient to 10^{-16} (Højsgaard et al., 2006). Sandwich derivatives use numDeriv (Gilbert and Varadhan, 2019). The production run is `simulation/run_main.R` under R 4.3.3 (R Core Team, 2024), with the `survey` package version 4.2.1 (Lumley, 2024).

lavaan 0.6.17 and OpenMx 2.21.11 were loaded by the replication preflight and were not used to fit these models. Koziol's models are weighted survey regressions and weighted linear mixed models, not structural equation models, and lavaan does not implement Mplus multilevel pseudo-maximum likelihood with separate level-1 and level-2 sampling weights.

4 Assumptions

4.1 The normal outcome is the latent response in Appendix A

Appendix A always draws a continuous $Y^\star$ and a binary threshold. The normal condition analyzes $Y^\star$. We did not analyze the binary variable as if it were the normal outcome.

4.2 Level-1 covariate centering uses the population cluster mean

The saved analysis columns in Appendix A include the population group-mean-centered covariate, not the sample cluster mean. We centered the same way, before sampling.

4.3 Method 2 scaling is applied before the pseudo-likelihood

Mplus `WTSCALE=CLUSTER` is Pfeffermann Method A, called Method 2 in the dissertation, and level-2 weights are unscaled. WeMix does not scale weights itself. We scaled, then passed conditional level-1 weights and unscaled level-2 weights. The agreement with WeMix is agreement after that scaling, not evidence that an unscaled weight would have matched Mplus.

4.4 Intervals use 1.96

Coverage counts a parameter as covered when it lies inside the estimate plus or minus 1.96 standard errors. That is a large-sample Wald interval. It is not the t critical value that `svyglm` would use for a confidence interval at the design degrees of freedom.

4.5 Each replication has its own seed

Appendix A uses one `set.seed` and then 1,000 sequential draws, and it draws the informative and non-informative samples in the same loop. We used seed $1000 \times \text{cell} + \text{replication}$ for one design at a time. The Monte Carlo distribution is the same. The individual replicate files are not.

4.6 Unweighted multilevel standard errors are model-based

Mplus computed MLR standard errors for every estimator. Our unweighted multilevel standard errors are the model-based maximum likelihood standard errors from `lmer`. Under informative sampling the intercept bias is so large that coverage is 0 either way. Under non-informative sampling those model-based intervals still covered at rates between 0.945 and 0.960, inside or beside the band she treated as consistent with .95.

4.7 Variance components use the Appendix A construction

Table 3.2 rounds τ_{00} to 0.053 at intraclass correlation .05. Appendix A uses 0.052632, which is .05/.95. We used $\tau_{00} = \rho/(1-\rho)$ and set each fixed slope so that the predictor accounts for six percent of the relevant variance, which reproduces the extra digits in Appendix A. Targets in the bias formulas are those generation values, not the three-decimal table.

5 Ambiguities that were resolved

5.1 Which file the normal analysis reads

The generator writes both $Y^\star$ and the binary Y. The normal model has an identity link and a free residual variance, so the analyzed column is $Y^\star$. Using the binary column would have produced a different intercept scale. The normal results line up with Table C.2, which is the check.

5.2 How the approximate cluster weight is counted

Equation (16) is M/m. The code divides by the number of clusters that remain after level-1 selection, not by the number of clusters that passed the stage-1 draw and then lost every student. We used that

realized count. The approximate-weight intercept bias of 0.309 against 0.308, and 0.440 against 0.440 at the large-cluster high-correlation cell, is the check.

5.3 Whether a second package reproduces the weighted multilevel point estimate

WeMix was fit to the same scaled weights as the analytic pseudo-likelihood. Coefficients agreed to about 10^{-6} and standard errors to about 10^{-4} on the production seed for replication 1. That closes the candidate that the analytic integral was a different functional from the package that implements multilevel pseudo-maximum likelihood. It does not by itself prove that Mplus 7 used the identical finite-sample sandwich.

5.4 Whether the single-level weight is the unconditional weight

The survey estimator and the independence GEE, both with the product of the conditional and cluster weights, returned the same coefficient to machine precision on one informative draw. The point estimate is weighted least squares. The small standard-error difference between the two sandwiches did not change the coverage story at the primary cell, where coverage is 0.945.

6 Comparison of published and replicated values

Numerical agreement is judged at the printed precision. A match rounds half up to the printed value. Close means it does not, and the absolute gap is at most one unit in the last printed place. Anything larger is a mismatch. Scope is limited throughout: the procedure was reconstructed in R, the seeds are not hers, and the Mplus input files were not obtained. Two-standard-error bands below use our Monte Carlo standard error of the mean and treat the printed number as exact unless the text says otherwise. With eight informative intercept-bias contrasts for the three weighted estimators, a Bonferroni reference on those 24 comparisons would be very wide; the bands below are unadjusted, and they are not equivalence tests.

Table 3: Selected comparisons. Precision is the printed unit. The contrast rows use the difference of the two printed biases.

Claim	Published	Ours	Precision	Numerical	Scope
Primary intercept bias, ws.ics	0.000	0.001	0.001	close	limited
Primary intercept bias, wtm.ccs	0.151	0.153	0.001	mismatch	limited
Primary intercept bias, wam.ccs	0.308	0.309	0.001	close	limited
Primary contrast, wtm minus ws	0.151	0.152	0.001	close	limited
Primary contrast, wam minus wtm	0.157	0.156	0.001	close	limited
Primary coverage, ws.ics	0.946	0.945	0.001	close	limited
Primary coverage, wtm.ccs	0.001	0.000	0.001	close	limited
$\bar{n}_j = 20$, $\rho = .25$, wam.ccs bias	0.440	0.440	0.001	match	limited
$\bar{n}_j = 20$, $\rho = .05$, wtm.ccs coverage	0.282	0.248	0.001	mismatch	limited

The primary true-weight multilevel bias, 0.153 against 0.151, is a mismatch at the third decimal and is about two of our standard errors above the printed 0.151 if that print is treated as exact. The contrast of that bias with the single-level bias is close at one thousandth. The coverage mismatch at $\bar{n}_j = 20$ and $\rho = .05$ is described above. None of these gaps reverses the order of the estimators.

7 What remains unresolved

The Bernoulli arm and the random-coefficient arm were not fit. Her informative Bernoulli random-intercept cell at $\rho = .05$ and $\bar{n}_j = 5$ reports a weighted single-level intercept bias of 0.000 and a true-weight multilevel bias of -0.170, with the level-1 slope relative bias 0.124 for the multilevel estimator and 0.000 for the single-level estimator (Table C.2). A probit pseudo-likelihood with 21 quadrature nodes was started on one such sample and did not converge inside the iteration limit, so that draw is not evidence about the cell. WeMix 4.0.3 fits binomial models with the logit link, not the probit link she requested with `LINK=PROBIT`. Those 200 settings are blocked, not a failure to match a number we computed.

The Mplus 7 input files, the replicate files named in Appendix A, and the seed stream that produced Tables C–F were not in the dissertation PDF, the 2017 supplement (Koziol et al., 2017), GitHub, the OSF query we ran, or Zenodo. The ledger is `notes/availability.md`. Without those files the one-thousandth gaps cannot be split into Monte Carlo error, a sandwich scaling, and a generator detail we still have wrong. The candidate that moves the primary bias cell is ordinary Monte Carlo error plus the difference between CR-0 and Mplus MLR. The candidate that moves the $\bar{n}_j = 20$, $\rho = .05$ coverage cell is the same sandwich scaling: a five percent larger standard error reproduces a coverage of 0.288, next to her 0.282. The artifact that would settle both is a Mplus 7 output, or the saved replicate estimates, for those two cells.

Forty-seven replications in the informative large-cluster cells, and fewer elsewhere, were dropped because `nlminb` did not return convergence code 0 or a standard error was missing. The finite coefficient on those draws was not extreme. Dropping them changes the third decimal of coverage in the sensitive cell by about 0.004. That is not the 0.034 gap.

8 Code and source availability

The dissertation PDF used here is the Internet Archive copy of the DigitalCommons file, stored as `source/koziol2015.pdf`. Direct download from DigitalCommons returned a Cloudflare challenge on 2026-11-02. Appendix A is the only program text in the dissertation. The 2017 supplement repeats tables and does not add syntax. Places checked, the UTC date, and the result are in `notes/availability.md`.

9 Next steps

The next check is to refit two cells in Mplus 7, or to read saved Mplus estimates: the informative normal random-intercept cell at $\rho = .05$ and $\bar{n}_j = 5$, and the same cell at $\bar{n}_j = 20$. The comparison is the intercept point estimate and the MLR standard error on the same samples our generator writes. A second check, if that file is available, is the informative Bernoulli probit at $\bar{n}_j = 5$, which is the arm her discussion treats as the harder case for multilevel weights.

10 Compliance ledger

Requirement	Status	Evidence
Fit in R	followed	R 4.3.3. Production script `simulation/run_main.R`.
lavaan with OpenMx as second optimizer	departed	Preflight loaded lavaan 0.6.17 and OpenMx 2.21.11. The dissertation does not fit a structural equation model. Weighted multilevel pseudo-maximum likelihood was fit analytically and checked against WeMix on a shared sample.
Second implementation of the weighted estimators	followed	WeMix for the multilevel pseudo-likelihood; geepack for the weighted single-level coefficient.
Whole dissertation inventoried	followed	Chapters and appendices are in the evidence packet. Bernoulli and random-coefficient arms are blocked, not omitted.
Do not assert an author error	followed	Open gaps are stated as unresolved causes.

11 Artifacts

Each command is run from `formalization/koziol-weighted-multilevel/simulation` unless noted. `R_LIBS_USER` must contain WeMix, and `OMP_NUM_THREADS=1`.

Claim	File	Command
Generation and the eight cells	simulation/run_main.R	`SIM_REPS=1000 SIM_CORES=3 Rscript run_main.R`
Monte Carlo summary	simulation/results/summary.csv	`Rscript summarize.R`
WeMix check	simulation/results/wemix_crosscheck_cell1.csv	One cell1 mix on seed 1001
Python generator check	simulation/generate_py.py	`Rscript check_generators.R` then `generate_py.py` on the saved draws
Source PDF	source/koziol2015.pdf	Archive snapshot cited in the availability note

The cell files simulation/results/cell_01.csv through simulation/results/cell_08.csv hold every retained and dropped replicate. Rebuilding the PDF from the note's directory is `pdflatex`, `bibtex`, `pdflatex`, `pdflatex` on paper/weighted_single_level_versus_multilevel__a_replication_of_koziol_2015.t

References

Tihomir Asparouhov. General multi-level modeling with sampling weights. *Communications in Statistics – Theory and Methods*, 35:439–460, 2006. doi: 10.1080/03610920500476598. Bibliographic details as printed in Koziol's reference list.

Paul Bailey, Blue Webb, Claire Kelley, Trang Nguyen, and Huade Huo. *WeMix: Weighted Mixed-Effects Models Using Multilevel Pseudo Maximum Likelihood Estimation*, 2023. URL https://CRAN.R-project.org/package=WeMix. R package version 4.0.3.

Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1):1–48, 2015. doi: 10.18637/jss.v067.i01.

David A. Binder. On the variances of asymptotically normal estimators from complex surveys. *International Statistical Review*, 51:279–292, 1983. Bibliographic details as printed in Koziol's reference list.

Paul Gilbert and Ravi Varadhan. *numDeriv: Accurate Numerical Derivatives*, 2019. URL https://CRAN.R-project.org/package=numDeriv. R package version 2016.8-1.1.

Søren Højsgaard, Ulrich Halekoh, and Jun Yan. The R package geepack for generalized estimating equations. *Journal of Statistical Software*, 15(2):1–11, 2006.

Milorad S. Kovačević and Shesh N. Rai. A pseudo maximum likelihood approach to multilevel modelling of survey data. *Communications in Statistics – Theory and Methods*, 32:113–135, 2003. Title and year as used by Koziol for the cluster-weight approximation. Not re-read against the journal.

Natalie A. Koziol. *A Comparison of Population-Averaged and Cluster-Specific Approaches in the Context of Unequal Probabilities of Selection*. Doctoral dissertation, University of Nebraska–Lincoln, 2015. URL https://digitalcommons.unl.edu/cehsdiss/229/.

Natalie A. Koziol, James A. Bovaird, and Sonia Suarez. A comparison of population-averaged and cluster-specific approaches in the context of unequal probabilities of selection. *Multivariate Behavioral Research*, 2017. doi: 10.1080/00273171.2017.1292115. DOI and author list taken from the Figshare record 10.6084/m9.figshare.4743673. The article PDF was not re-read for this note.

Thomas Lumley. Analysis of complex survey samples. *Journal of Statistical Software*, 9(1):1–19, 2004.

Thomas Lumley. *survey: Analysis of Complex Survey Samples*, 2024. URL https://CRAN.R-project.org/package=survey. R package version 4.2.1.

Linda K. Muthén and Bengt O. Muthén. *Mplus User's Guide*. Los Angeles, CA, 7 edition, 2012. Version named by Koziol on dissertation page 76.

Danny Pfeffermann, C. J. Skinner, D. J. Holmes, Harvey Goldstein, and J. Rasbash. Weighting for unequal selection probabilities in multilevel models. *Journal of the Royal Statistical Society, Series B*, 60:23–40, 1998. Bibliographic details as printed in Koziol's reference list.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2024. URL https://www.R-project.org/. Version 4.3.3.

Sophia Rabe-Hesketh and Anders Skrondal. Multilevel modelling of complex survey data. *Journal of the Royal Statistical Society, Series A*, 169:805–827, 2006. Bibliographic details as printed in Koziol's reference list.